

Revised Generation Interconnection System Impact Study Report

For Queue Project Q12-0185

Paloma 100 kV
October 18, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0185. The Interconnection Customer proposes a 36.8 MW solar photovoltaic facility interconnecting at the Paloma 100 kV substation in the Illinois area. The request was received on December 10, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 120 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,102,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The ROYALTON-IVYGLEN MEER 230.0 kV is overloaded to 106.6% of its 521 MVA emergency rating for the loss of the FOT ST VRAIN-ALDER 500.0 kV. The project contributes 12.1% DFAX (4.5 MW) to this overload.

ALDER-SHADY OAKS WIND FARM 230.0 kV: 106.4% loading (810 MVA) following the outage of the SHADY OAKS WIND FARM SUBSTATION-ALDER SUBSTATION 500.0 kV; DFAX = 25.3%.

EASY VLY-HEFFLEY TAP 100.0 kV: 102.2% loading (110 MVA) following the outage of the WALDEN-W AMESBURY 100.0 kV; DFAX = 80.3%.

For the loss of DEKALB-RICHLAND 500.0 kV, the SEGS 5-HILLCREST 100.0 kV loads to 100.2% of its 157 MVA rating. Project DFAX: 7.5%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
IVYGLEN METER-ROYALTON 230.0 kV	Reconductor line	\$3,925,000	3% (\$108,000)
SHADY OAKS WIND FARM-ALDER 230.0 kV Line	Reconductor line	\$11,247,000	17% (\$1,939,000)
HEFFLEY TAP-EASY VLY 100.0 kV	Reconductor line	\$3,049,000	100% (\$3,049,000)
SEGS 5-HILLCREST 100.0 kV	Reconductor line	\$894,000	1% (\$6,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$5,102,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.